

Stage 8: Product Catalog & Multi-Level Category Hierarchy Specification

Executive Overview

Stage 8 delivers the multi-tenant product catalog and category taxonomy engine for the Alight International Multi-Vendor Marketplace. It powers self-referencing category trees, brand discovery, multi-variant SKU matrices, tax HSN/SAC classifications, and vendor-scoped product lifecycles.

Key Features & Architectural Highlights

1. Multi-Level Category Hierarchy

- Recursive Tree Structure:** Infinite nesting depth with self-referencing `parent_id` foreign keys and optimized ordering via `display_order ASC, name ASC`.
- Category Commission Overrides:** Categories can define specific commission rates (e.g. 6.00% on Smartphones, 12.00% on Jewelry) that override default vendor commission tiers.
- Hierarchical Tree API:** Single-call tree retrieval (`/api/v1/categories/tree`) designed for responsive mega-menus and mobile navigation drawers.

2. Brand Ecosystem

- Brand Management:** Dedicated brand registry with slugs, high-resolution logos, banners, official website links, and featured flags.
- Admin & Public APIs:** Administrative brand curation and public brand directory for storefront filtering.

3. Rich Multi-Variant SKU Engine

- Product Metadata:** Multi-language short and full Markdown descriptions, search tags, HSN/SAC tax codes, and low stock threshold alerts.
- Multi-Variant SKU Matrices:** Variants track individual SKU, variant name, barcode (UPC/EAN/GTIN), real-time price, compare-at price, weight in grams, dimensions, variant-specific image URL, and structured JSON attributes.
- Dynamic Attribute Specifications:** Extensible key-value attribute system (e.g. Material, Screen Size, Processor, Warranty) rendered with custom display ordering.
- Gallery Images:** Multiple image support with primary thumbnail designation and sort ordering.

4. Vendor Lifecycle & Admin Moderation

- Product States:** `DRAFT` `PENDING_APPROVAL` `APPROVED` (Published) / `REJECTED` / `ARCHIVED`.
- Multi-Tenant Security:** Vendors can only view, modify, and delete products belonging to their store.

- **Admin Review Queue:** Administrators can inspect submitted products, approve for live marketplace listing, or reject with a detailed correction notice.

Database Migrations (`v34__product_catalog_and_category_hierarchy.sql`)

1. **categories** Table: - `commission_percentage NUMERIC(5, 2)` 2. **brands** Table: - `is_featured BOOLEAN NOT NULL DEFAULT FALSE` 3. **products** Table: - `hsn_code VARCHAR(50)` - `tags TEXT` - `low_stock_threshold INT NOT NULL DEFAULT 5` 4. **product_variants** Table: - `barcode VARCHAR(100)` - `compare_at_price NUMERIC(12, 2)` - `weight_grams NUMERIC(10, 2)` - `image_url VARCHAR(512)`

API Surface

Module	Method	Endpoint	Description
Category	GET	<code>/api/v1/categories/tree</code>	Full public hierarchical tree
Category	GET	<code>/api/v1/categories/{parentId}/subcategories</code>	Get direct child subcategories
Category	GET	<code>/api/v1/categories/{slug}</code>	Public category detail by slug
Brand	GET	<code>/api/v1/brands</code>	List active marketplace brands
Brand	GET	<code>/api/v1/brands/{slug}</code>	Public brand storefront detail
Vendor Products	GET	<code>/api/v1/vendor/products</code>	Vendor's catalog with status filtering
Vendor Products	POST	<code>/api/v1/vendor/products</code>	Create product with variants & images
Vendor Products	PUT	<code>/api/v1/vendor/products/{id}</code>	Update product specifications & SKU
Vendor Products	PUT	<code>/api/v1/vendor/products/{id}/submit</code>	Submit for administrator review
Admin Products	PUT	<code>/api/v1/admin/products/{id}/status</code>	Approve or reject vendor product

Verification & Integration Tests

- Integration tests in `ProductCatalogIntegrationTest.java` .